Version Control Systems ? Git & Collaborative Workflows

RCS 305 ? Software Engineering | Riara University Revision Resource

1. Centralized vs. Distributed Version Control

Centralized VCS (e.g., SVN) relies on a single central server, creating a single point of failure and requiring continuous network connectivity. Distributed VCS (Git) grants every developer a complete repository clone with local history, branching, and cryptographic commit verification.

2. Core Git Mechanics & Data Objects

? Directed Acyclic Graph (DAG): Git represents project history as an immutable graph of commit nodes.
? The Three Trees: Working Directory (local files), Staging Area / Index (staged changes ready for commit), and Repository / HEAD (recorded commits).
? SHA-1 / SHA-256 Hashes: Every object (blob, tree, commit, tag) is addressed by its cryptographic digest, guaranteeing data integrity.

3. Enterprise Branching & Release Workflows

? Feature Branching: Developers isolate new features in dedicated branches (‘feature/login-jwt‘), merging via peer-reviewed Pull Requests (PRs).
? GitFlow: Structured release paradigm utilizing ‘main‘ (production-ready), ‘develop‘ (integration), ‘feature/*‘, ‘release/*‘, and ‘hotfix/*‘ branches.
? Merge Strategies: Fast-Forward, Three-Way Merge commit, and Rebase for linear history preservation.

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